



Identification of glaucoma in fundus images utilizing gray wolf optimization with deep convolutional neural network-based resnet50 model

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Abstract

The main objective of this research is to explore and predict the potentiality of the image analysis model for early detection and diagnosis of glaucoma for assessing ocular pathologies. The proposed Computer-Aided Diagnosis system can assist the ophthalmologist in diagnosing ocular diseases by providing a second opinion as a human expert's decision. The database images are acquired from the various public datasets available on the website and in the clinical. After that, these images are collected to perform the preprocessing techniques, which improve the quality of the image in their appearance and experimental results. Next, image segmentation is performed for blood vessel segmentation from the fundus retinal image. Then the segmented output is fed Gray Wolf Optimization with a deep convolutional neural network-based ResNet50 model. Ultimately, the Fundus images are classified by utilizing a Gray Wolf Optimization with deep convolutional neural network-based ResNet50 model-based transfer learning classifier. A different benchmark dataset and a private dataset are used to validate the performance of the proposed algorithm. The efficacy of the proposed system is evaluated by different performance measures like accuracy, sensitivity, and specificity. The results show the improvement of diagnosis in the proposed method compared to other methods.

Keywords Bilateral with Unsharp Filter · Glaucoma detection · Gray Wolf Optimization with deep convolutional neural network classifier · Hysteresis thresholding with morphological post segmentation process · ResNet-50

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1 Introduction

The primary focus of this research is to develop a CAD system for early detection and to aid in the screening and management of glaucoma [47]. Glaucoma is another leading cause of blindness worldwide and ranks third in India, which affects peripheral vision. Since the damage caused is irreversible and early detection of glaucoma is significant for preventing eye disease from getting severe, mass screening would be the only option for early detection among the immense population for predicting the severity of glaucoma. Fundus camera is the cheapest imaging analysis modality that suits the monetary demands of the population. The characterization of glaucoma can be done by extracting structural features from the segmented optic disc and optic cup. The term "glaucoma" refers to a series of eye diseases that gradually weaken the optic nerve and, if left untreated, can result in blindness. The second most frequent cause of blindness in the world is glaucoma [43, 48]. According to the World Health Organization (WHO), glaucoma causes blindness in 4.5 million individuals [44, 45]. The ophthalmologists who treat cataracts, glaucoma, and diabetic retinopathy must all be taught more, clinics must be expanded, and low-cost surgical procedures must be made available, according to VISION 2020 [46]. The total number of people affected by glaucoma is estimated to be 111.8 million by 2040 [49]. In India, the estimated number of cases of glaucoma is 1.2 million, which is around one-fifth of the global burden of glaucoma. Currently, some medications can stop eyesight loss, but the condition needs to be caught early. The most important aspect is that it won't cause any early symptoms. The disease's main contributing element is an increase in ocular pressure. Furthermore, glaucoma can develop in adults with normal eye pressure.

There are other forms of glaucoma, but the two that immediately narrow the field of vision are Primary Open Angle Glaucoma (POAG) and Angle Closure Glaucoma (ACG). Deep learning and machine learning technologies have made significant strides recently in the development of new algorithms for automatically diagnosing glaucoma. This study suggests a paradigm for glaucoma automated diagnosis that enhances glaucomatous change detection and predicts fast progressive glaucoma. For image enhancement, segmentation, feature extraction, and classification, the suggested method makes use of appropriate techniques. The output's accuracy can be verified using the right metrics. Recently many Computer-Aided Diagnosis (CAD) systems have been developed [47] to identify the early stages of glaucoma through fundus images. It is an important goal for an ophthalmologist because early detection of glaucoma is a key factor leading to successful treatments. CAD system involves the use of computers to bring suspicious areas of a fundus image to an ophthalmologist's attention for further evaluation. For screening a large number of patients, a CAD system along with automated pre-screening by computers is the potential way to solve many problems. Even if there is no large-scale screening, computerized image analysis can be used to improve the quality of glaucoma analysis. In this scenario, computerized medical image analysis is used to find the possible affected regions in the image so that the ophthalmologist can examine those regions carefully. The main objective of this research is to predict the potentiality of the image analysis model for early detection and diagnosis of glaucoma for assessing ocular pathologies. The proposed CAD system would assist the ophthalmologist in diagnosing ocular diseases by providing a second opinion as a human expert's decision. The approaches investigated in this research

work are summarized below: A two-stage segmentation and classification model for the detection and diagnosis of glaucoma based on a deep learning framework. The main contribution of this research work is as follows:

Stage 1: Obtain the eye fundus images taken from the Public ORIGA, SCES, and SINDI dataset [Online] and the in-house clinical dataset from Dr. Somervell Memorial C.S.I. Medical College & Hospital, Kerala, India [Offline]. Initially, the input images undergo preprocessing and enhancement using a Contrast Limited Adaptive Histogram Equalization. Then the segmentation and extraction of Retina Blood vessels are carried out by hysteresis thresholding in combination with the morphological post-segmentation process.

Stage 2: Then the segmented output is fed to the research Gray Wolf Optimization with deep convolutional neural network-based ResNet50 model-based transfer learning classifier. Finally, evaluate the performance of the proposed model using accuracy, sensitivity, and specificity and compare the proposed model with state-of-the-art methods.

2 Related works

Automated glaucoma detection has become a major challenge in the field of medical image processing, where the most important objective is to increase the glaucoma detection accuracy efficiently from the available image database. There are many existing methods available to detect glaucoma explained in the given literature. However, the conventional methods are less accurate for the diagnosis of glaucoma. Therefore new methods are required to detect glaucoma more accurately. Gang et al. [1] have proposed a technique in which a Gaussian filter is used for the detection of retinal vessels in fundus photographs. The Gaussian filter used here is amplitude modified second-order Gaussian filter. The advantage here is a measurement of vessel diameter involves the process of verifying the continuity of diameter measured if it varies drastically then the operation is stopped. Also, vessel width measurement provides both sizes of the blood vessel and is used for optimizing the filter to improve the detection.

[3] used the Principal Component Analysis (PCA) to correlate pixel values across three color bands to help eliminate the influence of blood vessels on the segmentation of OD (Optic Disc) and the results obtained an improved image of the vascular tree. The canny edge identification is then used to increase the distinction between vessels and non-vessels. Rodrigues et al. [4] utilized multiscale filtering based on hessian using a genetic algorithm to get the optimum segmentation parameters for blood vessels. Chai et al. [5] used a Fully Convolution Network (FCN) and collected the patient data and the contours of anatomical structure from case reports, and they were able to produce an image of the fundus. A two-stage framework for ROI (Region of Interest) localization and glaucoma classification was created by [42]. The first stage was completed in two steps as a result. In the beginning, a semi-automated ground truth (GT) was created, distinguished by an architecture that recognized the OD automatically. The classification process was then carried out. The deep network was used once the OD was retrieved. The architecture was then split into four convolutional layers, including a Max pooling

overlapping strides layer and three fully connected layers. The use of cross-validation produced the most effective outcomes.

[6] invented a hybrid strategy to identify the contours of OD and OC (Optic Cup), which is also adaptively regularized and combines kernel-based Fuzzy C-Means (FCM) with the level set method to differentiate between the two. Hasan et al. [2] present an article on end-to-end networks designed for segmentation and localization of OD and Fovea using Dissecting and Reconstructing the convolutional neural network (DRNet). The method used is based on circular area searching. However, to estimate the OD radius, the method requires information about the camera's range of view. Imtiaz et al. [9] divided the OD and OC using region-based convolutional neural networks (RCNN). The VGG16 algorithm is used to extract features first then the split of the resulting structure after missing features are recovered. Two parallel, divergent branches are used to convey information into both the disc network and the cup network. The system uses a disc attention module to interpret the results of the disc network and the cup network.

The use of deep learning (DL) algorithms [14], most recently, has improved the diagnostic precision of these automated glaucoma detection systems [10]. Recently, a large number of computer-based methods for automatically diagnosing glaucoma using retinal fundus images have been developed [36, 38, 11, 13, 12, 34]. In this research, we suggested a novel computer-aided diagnosis (CAD) system that uses a deep learning classifier to categorize the various stages of the fundus images i.e., normal case or glaucoma case. Based on the literature the inferences drawn are as follows, there is a lack of an effective pre-processing approach for accurate segmentation. Lack of automated segmentation techniques, that overcome under or over-segmentation due to the presence of low contrast between the boundaries. Finally, lack of a state-of-the-art approach that takes into clinical knowledge of glaucoma screening.

Research gap identification & motivation for the research work Recently, many researchers [40, 22, 39, 17, 33, 31, 19, 32, 35, 18, 24, 25, 41, 26] proposed deep learning-based automated methods for medical image analysis purpose. The detection and diagnosis of glaucoma diseases are limited by the blood vessel segmentation of retinal fundus images. Further, clinical decisions are mostly identified based on intuitions and the experience of the ophthalmologist. This is based on the informative data that is concealed in the database, which may result in unnecessary errors and biases. A system for automatically identifying the characteristics of these pathological disorders will be highly beneficial for identifying diseases in medical imaging. Thus the research work aims the develop of CAD system for glaucoma detection.

3 Methodology

The Proposed method aims at early detection and diagnosis of glaucoma using a deep learning framework. The proposed algorithm has a two-stage network performing both segmentation and classification. In the first stage, the segmentation model is trained with the region of interest for the segmentation of the blood vessel. In the

second stage, a deep learning process is carried out from the segmented image based on a proposed method for the classification of the fundus images into normal and abnormal.

In this research work, both clinical and publically available database images [7, 30] were used, since they contain both Glaucoma and normal images. The Public ORIGA Database [35, 30] consists of a total of 650 fundus images. Singapore Chinese Eye Study (SCES) dataset [35, 30]; which consists of 1722 images. The SINDI dataset [35, 30] is a population-based study conducted also by the Singapore Eye Research Institute, aiming to assess the risk factors of visual impairment in the Singapore Indian community. In this dataset, there are a total of 5783 eye images. Then the in-house clinical database consists of 191 glaucomatous eye images obtained from Dr. Somervell Memorial C.S.I. Medical College & Hospital, Kerala, India.

The input fundus image is pre-processed using Contrast Limited Adaptive Histogram Equalization (CLAHE) for better image clarity [23]. Contrast enhancement is a ubiquitous term in fundus imaging to improve the quality of the images. CLAHE is a modified version of adaptive histogram equalization that prevents noise amplification and minimizes edge shadowing. The enhancement procedure in AHE is performed to a specific location, but in CLAHE, the entire image is split into tiny parts called tiles, and the contrast is boosted for each tile. The region may be improved by changing factors such as the number of tiles, the number of bins, the clip limit, and so on. The number of tiles (rectangular contextual areas) is set to [16 16], which is a positive integer 2-element vector. A contrast factor of 0.05 was utilized as a clip limit to prevent the image from being oversaturated.

Segmentation is the method of separating an image into various parts and the aim is to adjust the image representation and simplifies the image that is more coherent and easier for analysis. Image segmentation plays an essential role in the analysis of Fundus database images as the segmentation of the retina blood vessel [27] & extraction provides a vital sign for precise diagnosis of Glaucoma diseases. Moreover, segmenting the blood vessels [28] in fundus image plays a vital role in the diagnosis of various pathological disorders of an eye. For the objective of segmenting retinal blood vessels, hysteresis thresholding in combination with morphological post-segmentation is used in this research project. Using grey limit values within the thresholding intensity of the image, hysteresis thresholding is a bi-folded thresholding technique. In a few instances where object depth is high in some locations but falls below the noise level in other locations, the bi-folded thresholding method outperforms the single thresholding technique. A percentile-based threshold rule speeds up threshold computation [29]. To build up this rule, factors taken into consideration are, i. High assertive vessels were separated by the strong threshold, ii. High assertive background pixels were separated by a weak threshold. iii. The vessels within the map are classified through gray levels thicker than the background pixels. A strong threshold (T_s) is provided by the percentage that surrounds it. The background-enclosed image surface corresponds to the scrambled area integrated under the histogram. Therefore, the weak threshold (T_w) is calculated using deducting the part of the image surface positively enclosed using the background from 100%. Later T_s selects only vessel pixels and (T_w) chooses all vessel pixels practically.

In this research Gray Wolf Optimization with deep convolutional neural network-based ResNet50 model-based transfer learning classifier has been used for

the classification of retinal images. Deep learning has risen to the top of the machine learning hierarchy in recent years. Conventional deep learning architectures, such as Convolution Neural Networks (ConvNet), are ideally suited for analyzing medical images because of their ability to learn from previous experience. ConvNet is gaining popularity as a consequence of its ability to analyze and extract information from a wide range of visual sources. After the convolutional neural network has gained popularity there comes a need for optimizing the features in the image before it could be used for training convolutional neural networks. The general block diagram of the proposed works is given in Fig. 1.

Nowadays deep learning has gained popularity in image processing. Among the various types of deep learning techniques, the Convolutional neural network (CNN) has been commonly used for medical image analysis. A typical ConvNet design consists of a collection of

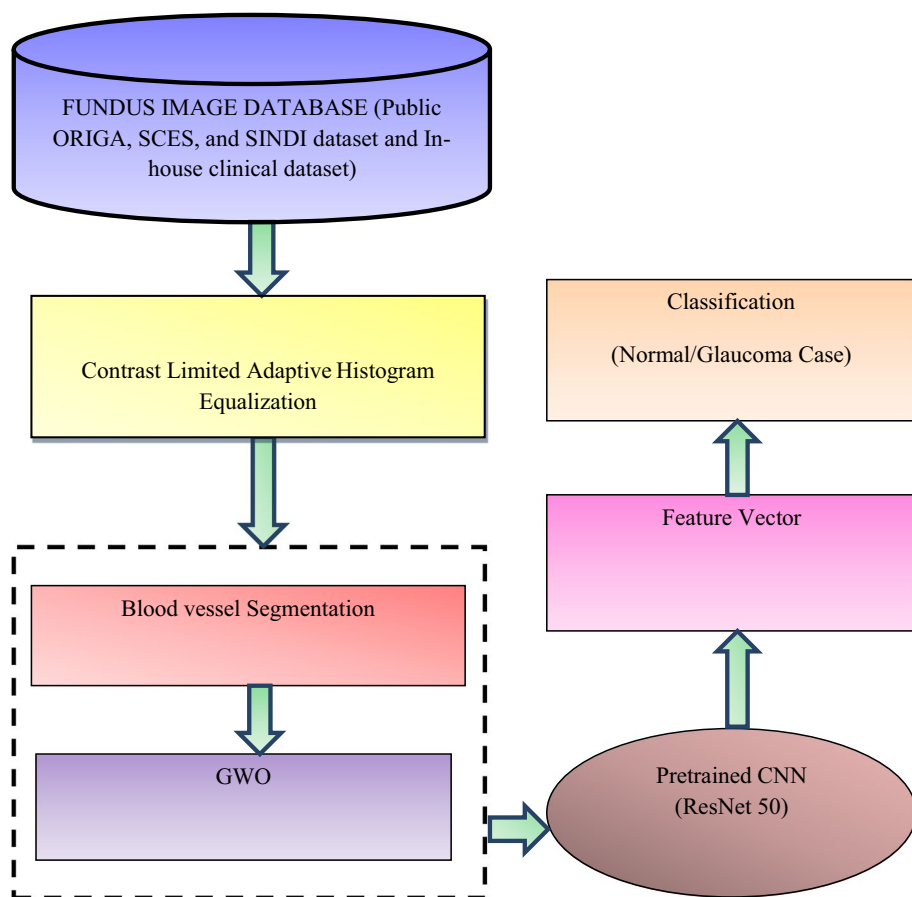


Fig. 1 Proposed Architecture of Gray Wolf Optimization with deep convolutional neural network-based ResNet50 model

convolution layers that are followed by an image pooling layer that is used to categorize the images being processed. It is made up of the Rectified Linear Unit (ReLU) network and the fully connected layer, and both are extensively employed in convolution and pooling layers to activate the neurons in the network. Different qualities may be obtained from feature maps by using convolution filters in conjunction with each other. This layer computes the maximum or average adjacent value for each feature map using the maximum or average neighboring value. As a result of this, increased process speed and processing time, and also the file size is reduced. Some information is lost as a result of this layer. So, the matrices are created by merging multiple layers of convolutional and pooling operations into a single softmax layer, which is then combined with further layers of convolutional and pooling processes. The output is produced by the softmax layer, which senses the parameters and weights of the applied matrices and creates them.

CNN outperforms all the other classification techniques mainly in terms of the processing speed because it does not involve any separate step for feature extraction and the CNN network performs on its own which helps in reducing the computation time [16]. An important class of machine learning is deep learning, which mainly uses the concept of the most popularly known as artificial intelligence. The reason for naming it artificial intelligence deep learning is that the network can automatically learn the feature values needed for classification instead of depending on the features extracted manually. There also comes a new way of deep learning that's transfer learning. Transfer learning involves using the knowledge gained by a pre-trained model that is already trained for some other problem. Thus by using the concept of transfer learning in the pre-trained networks learning can be done quickly instead of learning from scratch.

The main attraction of transfer learning is that a wide range of already extracted features can be used for our problem and also it can be made available even for a small number of datasets [20, 8]. When it comes to the categorization of fundus images, the suggested ConvNet model performs in a manner that is comparable to that of clinical trials. The early layers of ConvNet architecture often consist of hundreds of feature maps, which are arranged hierarchically. The number of feature maps diminishes gradually as one progresses through the levels. Each layer is made up of thousands of neurons, and the feature maps shown below are slices across each of those layers. ConvNet is a neural network generator that takes a single vector matrix as input and applies it to a group of neurons, which are referred to as hidden layers in the industry. When using classification settings, the output layer is completely linked, and the class scores are represented by the scores on the output layer.

The main aim of the proposed classification technique is to provide the optimized input to the pre-trained CNN classifier thereby increasing the extent of accuracy. Gray wolf optimization used here mainly deals with reducing the unwanted features in the input image and making it optimized for efficient classification by CNN. The Grey Wolf Optimization (GWO) algorithm [37] filters the redundant and irrelevant information by searching for the optimal value. The selection operation is performed by considering the fitness value, and it is calculated and sorted from the minimize function. Then they followed packs were created arbitrarily to replace the emigration wolf. The optimal solution is improved by applying the GWO [23] algorithm for optimization. The Gray wolf optimization algorithm mainly involves three main parameters α , β , γ which are considered as wolves by the gray wolf optimization algorithm. The GWO flowchart is shown in Fig. 2. The optimized image thus obtained is loaded into the pre-trained CNN for classification.

ALGORITHM STEPS:

Step 1: Initialize GWO parameters as GWi for $i = 1:n$ //****Parameter such as b, B, D and Imax

Initialization process****//

Initialize vectors b, B, D

$$B = 2 * b * r1 - b$$

$$D = 2 * r2$$

Initialize maximum number of iteration (Imax).

Step 2: Fitness values are estimated for each agents as follows//****Estimation of fitness value E

****//

$$E = |D * GWP(t) - GW(t)|$$

$$GW(t + 1) = GWP(t) - B * E$$

Step 3: Exploring the best agents as follows

//****Best Agents $E\alpha$, $E\beta$, $E\gamma$ and $GW1$, $GW2$, $GW3$ Exploring Process****//

$$E\alpha = |D1 * GW\alpha - GW|$$

$$E\beta = |D2 * GW\beta - GW|$$

$$E\gamma = |D3 * GW\gamma - GW|$$

$$GW1 = |GW\alpha - B1 * E\alpha|$$

$$GW2 = |GW\beta - B2 * E\beta|$$

$$GW3 = |GW\alpha - B3 * E\gamma|$$

Step 4: Update the current location of the agent //****Current location $GW(t + 1)$ updating process****//

$$GW(t + 1) = (GW1 + GW2 + GW3)/3$$

Step 5: Fitness value is estimated for all hunts.//****Process of estimating hunts fitness value****//

Step 6: Update the value of $GW1$, $GW2$ and $GW3$.//**** $GW1$, $GW2$ and $GW3$ values updating process****//

Step 7: If iteration reaches the maximum value then output the results otherwise move to step 4.

In the proposed work pre-trained network named Resnet 50 has been used for classification, being an automatic feature extractor it extracts all the possible number of features automatically and there exists no need for a separate feature extraction step. ResNet-50 requires input dimensions of 224x224 pixels. In this case, the input image is resized to 224x224 after it has been enhanced for contrast. A large amount of data is required to

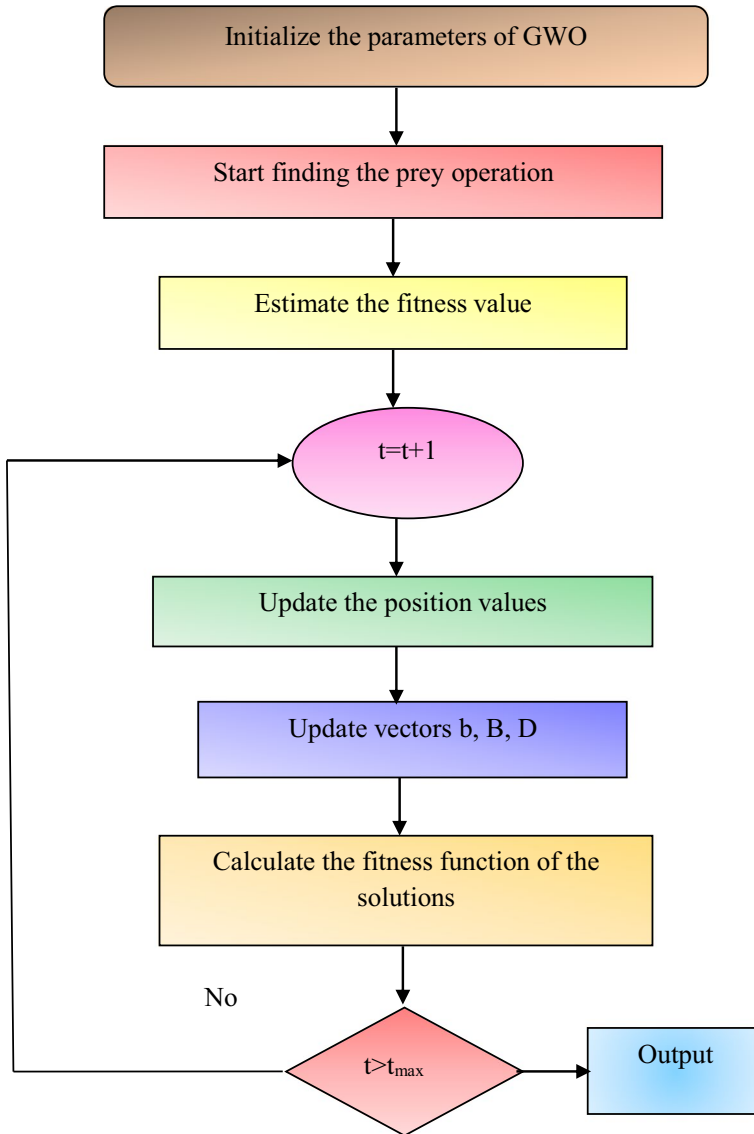


Fig. 2 GWO flowchart

achieve good efficiency with deep learning architectures. To develop an effective classifier, comprehensive training data are essential to improve the efficiency of a deep network. The data augmentation process involves a variety of techniques such as rotational transformation, and horizontal, and vertical flipping. In addition to the above, 6955 images are used in the added data collection, i.e. 1391 images are flipped vertically with a total of 3-turn transformations.

The Residual Network is a classical neural network architecture used to solve vanishing gradient problems. A repeated multiplication results in a very small gradient, in

Table 1 ConvNet configuration of ResNet-50

Layer Name	ResNet-50	Output size
Conv 1	7×7, 64, stride 2	112×112
Conv 2	3×3 max pool, stride 2 1 × 1, 64 [3 × 3, 64]×3 1 × 1, 256	56×56
Conv 3	1 × 1, 128 [3 × 3, 128]×4 1 × 1, 512	28×28
Conv 4	1 × 1, 256 [3 × 3, 256]×6 1 × 1, 1024	14×14
Conv 5	1 × 1, 64 [3 × 3, 64]×3 1 × 1, 2048	7×7
Final	Average pool 1000-fully connected Softmax	1×1

performance saturation and degradation during backpropagation. Shibata et al. [21] introduced ResNet which allows the model to learn an identity function that facilitates better performance of the higher layer. A residual network is a deep convolution network that consists of five stages each with a convolution layer. Feedforward neural networks with shortcut connections or bottleneck architecture can be used to formulate the output block. A convolution of size 7×7 and 64 different kernels of size 2 are provided as input in the first layer. There is also a max-pooling layer with a stride size of 2 and a three-time convolution layer. Table 1 shows a detailed description of the Resnet-50 architecture.

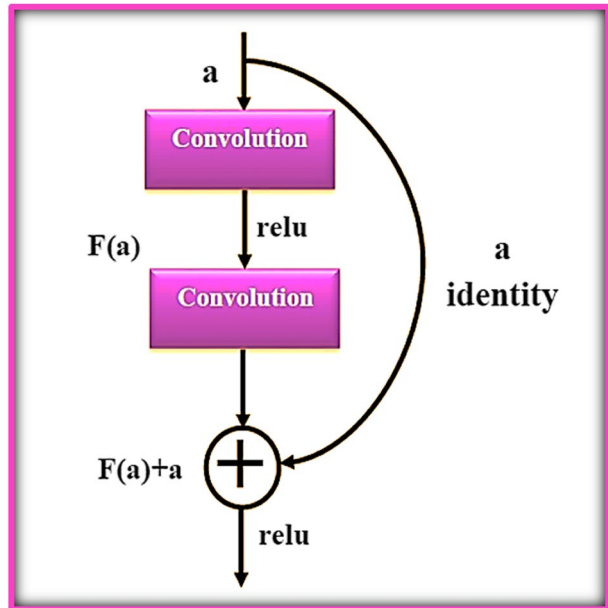
A few convolutional layers, batch normalizations, ReLu activation function, and one shortcut makes up the ResNet-50. The deep residual learning framework consists of function $f(a)$ and $b=a$ as identity mapping. The input to the block is added to the output block $F(a)$ as shown in Equation (1).

$$f(a) = F(a) + a \quad (1)$$

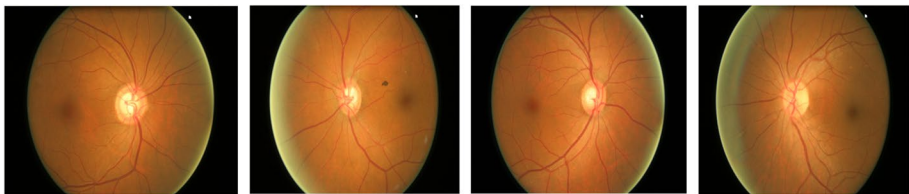
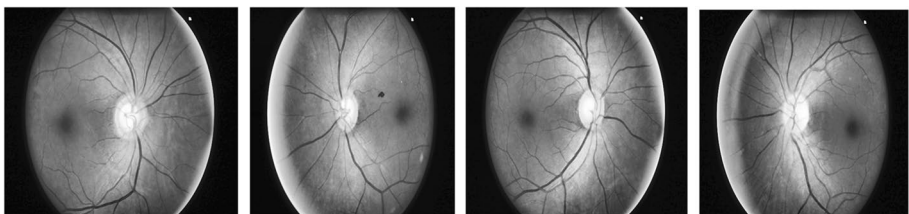
The shortcut connections perform identity mapping simply, and their outputs are added to the outputs of the stacked layers which is shown in Fig. 3. These shortcuts are useful for optimizing the trainable parameters to improve the final performance of the network.

4 Result & discussion

A proposed framework for glaucoma detection is presented in this section. Three benchmark datasets and one private dataset are used to test the performance of the proposed system. Datasets are divided into two parts: testing parts and training parts. The proposed model is implemented with MATLAB 2022a software in an i7 processor. Images of the

Fig. 3 Building Block of Residual Learning

retinal fundus in four sample glaucoma and healthy instances are shown in Figs. 4, 8. Pre-processing methods are employed to improve the image's quality or to rid it of noise. The imaging modality determines the source of noise in medical images, and filtering methods are selected based on the source of noise. There isn't a single filtering algorithm that applies to all medical images. Figures 5, 9 shows the pre-processed images of the Glaucoma and normal case.

**Fig. 4** Sample 4 input retinal fundus Glaucoma Case: Input image**Fig. 5** Sample 4 input retinal fundus Glaucoma Case: Preprocessed image

Figures 6, 7, 10, 11 show the segmented images of the Glaucoma and normal case retinal database samples. Finally, Retinal blood vessels are to be extracted for analysis of Glaucoma or normal case analysis. This segmented image feeds to the Gray Wolf Optimization with deep convolutional neural network-based ResNet50 model-based transfer learning classifier to classify the fundus images into the Glaucoma case and normal case image. Performance measures for deep learning architecture are evaluated using metrics such as accuracy, specificity, and sensitivity. The following equations provide the mathematical expressions of these parameters:

Sensitivity It is defined as the percentage of true detection made for abnormalities by the proposed framework. Sensitivity is given as

$$\text{Sensitivity} = \frac{TP}{TP + FN} \times 100\% \quad (2)$$

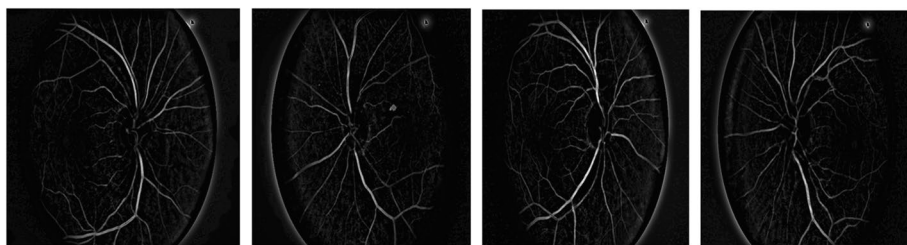


Fig. 6 Sample 4 input retinal fundus Glaucoma Case: Thick Vessel Image

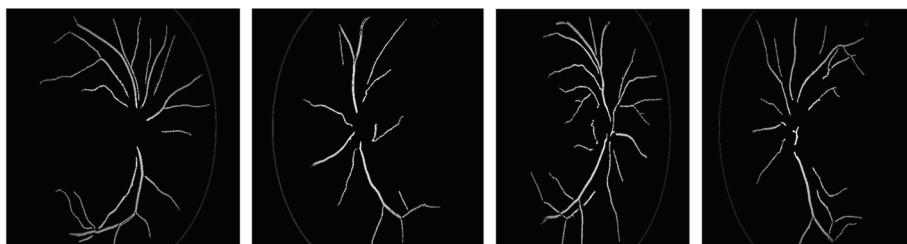


Fig. 7 Sample 4 input retinal fundus Glaucoma Case: Blood Vessel Segmented image



Fig. 8 Sample 4 retinal fundus Normal Case: Input image

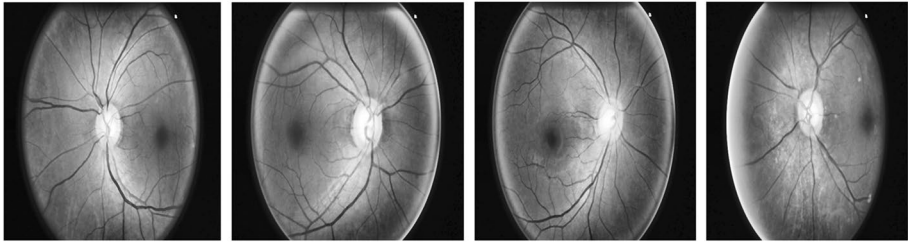


Fig. 9 Sample 4 retinal fundus Normal Case: Preprocessed image

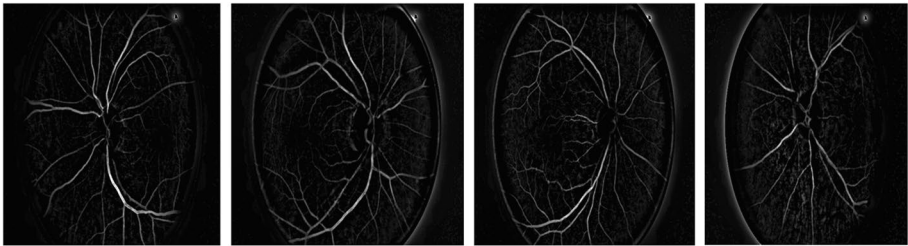


Fig. 10 Sample 4 retinal fundus Normal Case: Thick Vessel Image

Specificity It is defined as the percentage of true detection made for normal images by the proposed framework. Specificity is given as

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \times 100\% \quad (3)$$

Accuracy Accuracy is the measure for showing both normal and abnormal accurate detection by the proposed system. Accuracy is given as

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{FN} + \text{TN}} \times 100\% \quad (4)$$

Where the number of positive cases predicted correctly is TP, the number of negative cases predicted correctly is TN, the number of cases predicted incorrectly is FN when the case is negative, and the number of cases predicted incorrectly is FP when the case is positive.

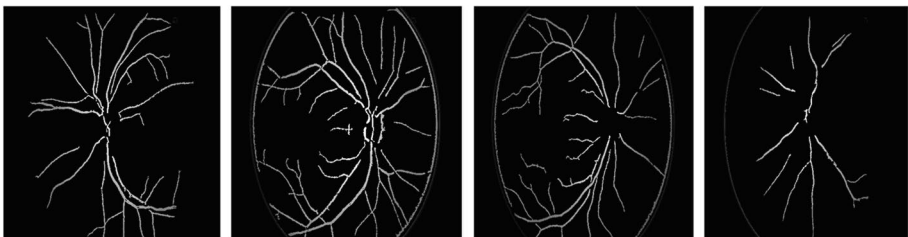


Fig. 11 Sample 4 retinal fundus Normal Case: Blood Vessel Segmented image

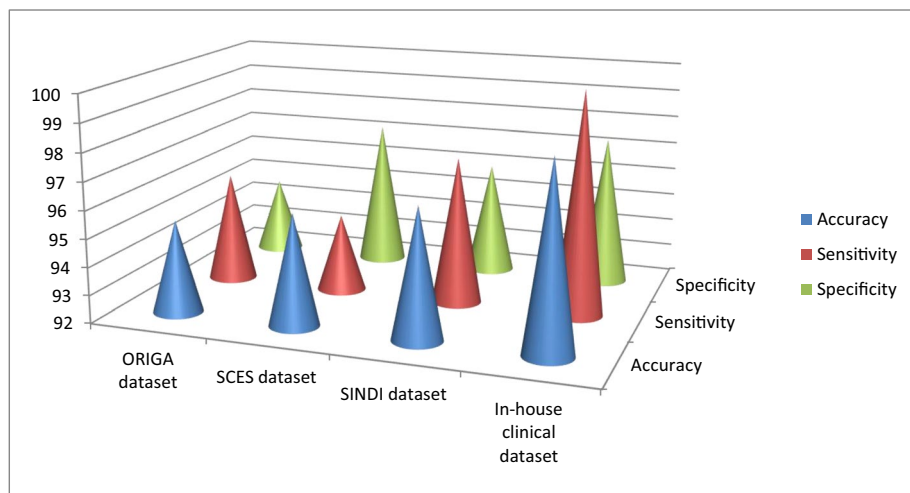


Fig. 12 Performance Plot for Gray Wolf Optimization with deep convolutional neural network-based ResNet50 model based transfer learning classifier

To evaluate the classification performance, majority voting strategies have been used. For this strategy, the entire dataset was initially divided into training and testing. In this study, 70% of the available data is utilized for training, with the remaining 30% used for testing. Although the training phase of Gray Wolf Optimization with deep convolutional neural network-based ResNet50 model-based transfer learning classifier models is time-demanding in general, it is not critical in classification issues since the classifiers are learned offline. Tables 2 and 3, Fig. 12 shows the Performance Plot for various parameter

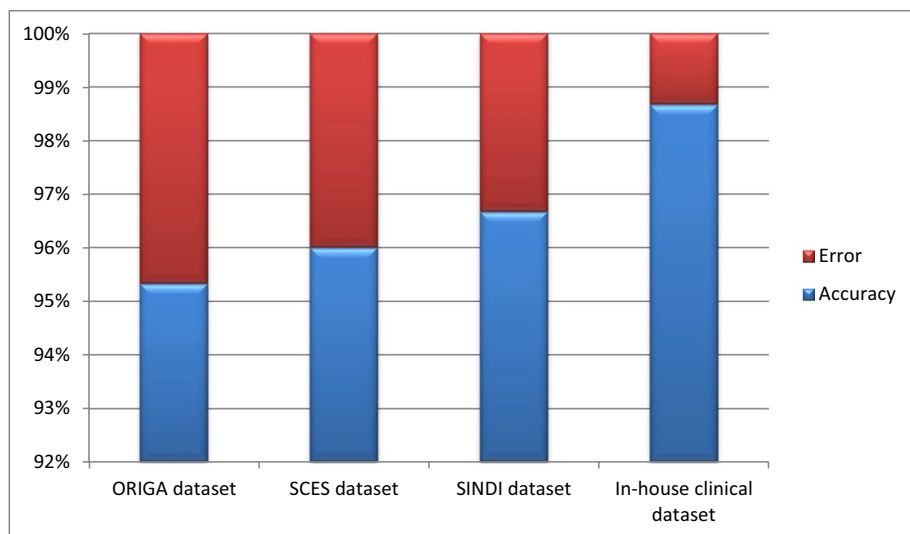


Fig. 13 Accuracy Vs Error Plot for Gray Wolf Optimization with deep convolutional neural network-based ResNet50 model based transfer learning classifier with different dataset

Table 2 Performance Comparison of proposed method with different dataset

Method	Dataset	Accuracy	Sensitivity	Specificity
Proposed Method	ORIGA dataset	95.33	95.94	94.73
(Gray Wolf Optimization with deep convolutional neural network-based ResNet50 Model based transfer learning classifier)	SCES dataset	96.00	94.80	97.26
	SINDI dataset	96.67	97.29	96.05
	In-house clinical dataset	98.67	100	97.40

Table 3 Comparison of Performance of Proposed Method with other deep learning method

Method	Dataset	Accuracy	Sensitivity	Specificity
ResNet	ORIGA dataset	86.86	81.44	91.93
	SCES dataset	87.04	82.08	91.54
	SINDI dataset	88.60	83.72	93.03
	In-house clinical dataset	91.11	86.55	95.20
VGG	ORIGA dataset	90.90	89.37	92.22
	SCES dataset	91.63	91.12	92.05
	SINDI dataset	92.72	91.67	93.62
	In-house clinical dataset	95.63	94.84	96.30
Proposed Method	ORIGA dataset	95.33	95.94	94.73
(Gray Wolf Optimization with deep convolutional neural network-based ResNet50 model based transfer learning classifier)	SCES dataset	96.00	94.80	97.26
	SINDI dataset	96.67	97.29	96.05
	In-house clinical dataset	98.67	100	97.40

Table 4 Performance accuracy comparison of various algorithms for Glaucoma detection

References	Methods	Performance
Proposed Method	(Gray Wolf Optimization with deep convolutional neural network-based ResNet50 model based transfer learning classifier)	Accuracy of 95.33% (ORIGA dataset) Accuracy of 96.00% (SCES dataset) Accuracy of 96.67% (SINDI dataset) Accuracy of 98.67% (In-house clinical dataset)
[13]	Deep learning based classification	Accuracy of 93.5%
[14]	Deep learning classifier	Accuracy of 93%
[15]	3D Convolutional neural network Classifier	Accuracy of 83.23%
[42]	Four convolutional layers and fully connected layers	Accuracy of 87.40%
[35]	Disc-aware ensemble network (DENet)	Accuracy of 91.83%
[5]	multi-neural network branch (MBNN) model	Accuracy of 91.51%

of the proposed Gray Wolf Optimization with deep convolutional neural network-based ResNet50 model-based transfer learning classifier with various dataset.

Figure 13 shows the Accuracy and error Plot for Gray Wolf Optimization with deep convolutional neural network-based ResNet50 model-based transfer learning classifier with different datasets. Table 4 shows the Experiments on both public and private datasets reveal that the proposed model outperforms convolutional neural network architecture and classic computer-aided diagnostic techniques.

Experiments on both public and private datasets reveal that the proposed model outperforms convolutional neural network architecture and classic computer-aided diagnostic techniques.

5 Conclusion & future scope of the research

In the proposed work, a novel model for the early diagnosis of glaucoma was developed. The automatic diagnosis of glaucoma from fundus images is a key component of the current investigation. Additionally, the suggested detection method is employed to confirm improved accuracy while avoiding complexity. The suggested Gray Wolf Optimization with deep convolutional neural network-based ResNet50 model-based transfer learning classifier from fundus images verifies improved accuracy and offers an effective detection at a higher level of sensitivity even with the growing number of images. With the use of MATLAB software and currently available automatic glaucoma identification on fundus images, a thorough simulation is conducted. In the future, the proposed research work can be implemented on an Android platform with multi-class glaucoma detection. This extended work can benefit people by detecting glaucoma at an early stage only on the Android platform at less cost. This will quickly and directly inform the stage of glaucoma and glaucoma-affected people may be cured by nearby eye care centers. In this way, it can reduce the chances of moderate or deep glaucoma and hence blindness in people. The accuracy of the proposed methodology may further be enhanced by training the large data set and better image pre-processing as well as decomposition techniques. In some cases, the poor image quality due to lighting effects during image acquisition may affect the pre-processing. It can be solved as a future problem.

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Authors' contributions Sujithra B S^{1*}: Roles: Conceptualization, Methodology, Validation, Visualization, Writing – original draft, Writing-Reviewer Comments Correction, Proof reading and Visualization

S.Albert Jerome 2: Roles: Visualization, Data Correction, Resources, and Validation Reviewer Comments Correction and Editing

Data availability The Code implementation, datasets generated during and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval This study has not been supported by any industrial company and does not serve to promote any commercial product. Anonymized publicly available databases were used in the conducted experiments.

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